AP PRECALCULUS

Unit 1 • Lesson 1.14 practice

Lesson 1.14 • Function Model Construction and Application

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

A rectangle has perimeter 40 cm and width w cm. Which function gives its area?

(A) $A(w) = w(40 - w)$
(B) $A(w) = w(20 - w)$
(C) $A(w) = 2w + 40$
(D) $A(w) = w^2 - 20$

Question 2

NO CALCULATOR

What is a reasonable domain for $A(w)$ in the previous question?

(A) $0 < w < 20$
(B) $0 < w < 40$
(C) $w > 0$
(D) All real numbers

Question 3

NO CALCULATOR

What is the maximum area of a rectangle with perimeter 40 cm?

(A) $400\ \text{cm}^2$
(B) $200\ \text{cm}^2$
(C) $100\ \text{cm}^2$
(D) $80\ \text{cm}^2$

Question 4

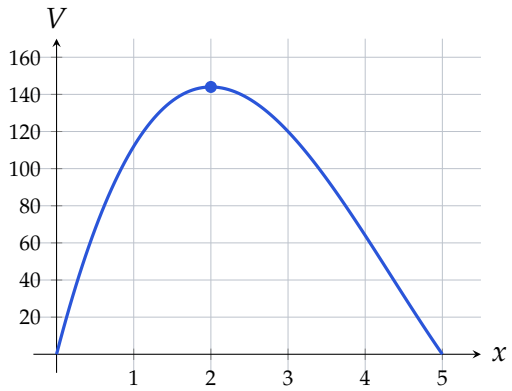
CALCULATOR

An open box is made from a 12 in by 12 in square sheet by cutting squares of side x from the corners. What is the maximum volume?

(A) $96\ \text{in}^3$
(B) $108\ \text{in}^3$
(C) $144\ \text{in}^3$
(D) $128\ \text{in}^3$

Question 5 (graph) NO CALCULATOR

The graph shows the volume $V(x)$, in cm^3, of an open box made from a 10 cm by 16 cm sheet with squares of side x cm cut from the corners. Which value of x gives the greatest volume?



(A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 5$

Question 6 (Free response) CALCULATOR

A cylindrical can must hold 500 cm^3. Let r be its radius and h its height, in centimeters.

(a) Express h in terms of r.
(b) Show that the surface area is $S(r) = 2\pi r^2 + \dfrac{1000}{r}$.
(c) Use a graphing calculator to find the radius that minimizes S, and the minimum surface area.
(d) Describe the behavior of S as $r \to 0^+$ and as $r \to \infty$, and explain why in context.